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Hard-CAD: Contract-Based Diagnostics for Generated CAD Geometry

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ABSTRACT

A CAD revision can improve global shape overlap while violating a previously satisfied local requirement. Hard-CAD checks both versions against the same explicit requirements, recording their entities, coordinate frames, states, tolerances and measured values.

We examine 41 eligible revision pairs and 84 paired cut requirements from 30 modeling briefs under two local agent conditions. In one revision, removing material from one void more than offsets a 28.27 mm^3 addition to a previously clear side-cut: intersection-over-union (IoU) improves and total extra volume falls, but the side-cut is newly occupied. In another, a void clears as total extra volume grows. Boolean differences locate the material added and removed in each edit. External checks produce comparable records; transfer and feedback trials examine repairs and new violations under specific conditions. Tool-using agents repair clearance, installation-path and travel violations. For real documents, we evaluate recovery of written dimensions, candidate geometry and product identifiers separately from feature-level semantic accuracy. The reports identify requirement changes, their magnitudes and unavailable measurements alongside aggregate scores.

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1. Introduction

A generated CAD model may preserve most occupied material while placing a hole on the wrong axis, blocking a specified void or fusing components that should remain distinct. An overlap score describes a global geometric relation; it does not specify which local requirements are satisfied. BenchCAD [1] documents a blocked through-bore at high voxel IoU, while Pointer-CAD v2 [2] evaluates dimensional fidelity through vertex, edge and face matching.

Reviewing a CAD revision requires checking both the intended repair and requirements that previously passed. Hard-CAD pairs the same requirement before and after an edit, reporting repairs, regressions and unchanged states alongside local occupancy and global material changes.

Each requirement specifies entities, frame, state, tolerance and source. Established Boolean, distance and prescribed-motion operations measure the geometry. Cohort references follow the supplied modeling recipes; document extraction is evaluated separately.

1.1. Contributions

1. **Contract-based revision records.** Paired records retain requirement identity, operands, frame, units, bound and the measured value or reason for unavailability (Sections 4–5).
2. **Analysis of generated revisions.** Across 30 briefs, we trace a new local violation during global improvement and a local repair during material growth, retaining acquisition failures and numerical exclusions (Section 6.4).
3. **Executable evaluation.** Replayable reports, binding controls, external checks and feedback trials test the workflow; a separate document audit evaluates input recovery (Sections 6.5–6.7).

In cad15, the revision removes 56.55 mm^3 from one declared void and adds 28.27 mm^3 to a previously clear side-cut. Total extra volume falls and IoU improves, while the side-cut becomes occupied. Figure 3 shows fixed-scope pairing and replay. Separate MiniMax trials cover clearance, installation order and travel.

2. Related Work

2.1. CAD Representations and Executable Generation

DeepCAD [3] and SkexGen [4] model CAD operation sequences. BRepNet [5] segments B-Rep faces through topological message passing; UV-Net [6] combines UV-mapped geometry with adjacency. BrepGen [7], AutoBrep [8] and DualBrep [9] generate B-Reps using diffusion, autoregression and joint flow matching, respectively. These representations expose different information to an evaluator. WHUCAD [10] spans sketch constraints, named topology and parametric history, while the Fusion 360 Gallery datasets provide design sequences [11] and assemblies with joints and contact surfaces [12].

Text2CAD [13] and CAD-MLLM [14] translate text into parametric CAD. CAD-Coder [15] generates CadQuery scripts with supervised learning and geometric/format rewards. CAD-Prompt [16] pairs natural-language prompts with expert-annotated code. Text-to-CAD Retrieval [17] retrieves candidates using procedural and point-cloud embeddings; Hard-CAD reports no retrieval-augmented generation experiment.

Execution and feedback are established signals. Query2CAD [18] generates and repairs CAD macros; CAD-Assistant [19] executes Python actions in FreeCAD; CAD-Recode [20] predicts Python from point clouds; CADFusion [21] uses visual feedback over parametric sequences. Hard-CAD also separates source execution from geometric measurements.

DeCAD [22] compares CAD states across operations and edits, using Boolean volume differences and reference visualizations to investigate reference errors. CADMorph [23] evaluates target-shape fidelity together with source-sequence preservation during parametric editing. Building on these uses of edit comparison and verification loops, our cohort analysis follows explicitly declared requirement states alongside material changes.

TraceCAD [24] combines persistent recovery state, localized candidate search and repair memory. Its paired regression measures distinguish lost reference-consistent geometry from newly introduced erroneous geometry. Our analysis relates such material changes to the identity and state of predeclared local requirements.

2.2. Requirement and Engineering Checks

SWE-bench [25] evaluates software repairs with executable tests. In CAD, CADTests [26] checks prompt-specified geometry and topology and uses test feedback to guide generation. CIT-CAD [27] infers a Constraint Intent Tree, checks generated programs against it, and accepts repairs only when they preserve satisfied constraints while reducing violations. Requirement checks and regression-aware repair are therefore established capabilities. Hard-CAD focuses on jointly recording requirement transitions, local occupied volumes and global material changes across a fixed revision cohort. CAD-Judge [28] uses compiler-based grading and review; MUSE [29] evaluates editable B-Rep assemblies through code, geometry and design-intent stages. BIM-Edit [30] checks consistency in edited building models, while Text2CAD-Bench [31] evaluates text-to-parametric-CAD generation. AssetOpsBench [32] and SOP-Bench [33] address industrial text workflows.

BenchCAD [1] evaluates executable CadQuery programs with geometric distances, voxel IoU, feature F1 and essential-operation recall, including paired edits and preservation checks. Its Appendix Q reports IoU 0.961 despite a blocked through-bore. Pointer-CAD v2 [2] evaluates vertex, edge and face matches at the original scale and location. Hard-CAD follows aggregate material changes and the same declared requirement across revisions (Figure 2). Table 1 compares documented roles, not performance across these different protocols.

CADGenBench [34] evaluates mating features through volumetric overlap with declared keep-in and keep-out regions, bounded pose search and weakest-feature aggregation. CADEngBench [35] combines parametric editing and design-for-manufacture checks with matched linear-static FEA, joint retrieval, geometric grounding and kinematic verification. HistCAD [36] records sketch primitives, constraints and feature operations for post-edit constraint checks. These works already evaluate local fit, engineering behaviour and constraint preservation.

AssemCAD [37] realizes and verifies typed parts, geometry-backed ports, mates and engineering axioms with kernel operations and traceability. Hard-CAD binds cohort requirements through explicit briefs and recipe-derived references; its geometry evidence does not depend on automatic document extraction or general component grounding. The optional *U* module checks declared labels.

Conventional CAD supplies minimum-clearance checking against user-supplied bounds, dynamic clearance, collision detection and static interference [38]. Hard-CAD uses these established classes of operations to retain interpretable requirement-level measurements across generated revisions.

2.3. Declared Actions and Evaluation Scope

Homem de Mello and Sanderson [39] give a correct and complete assembly-sequence generator based on contacts and precedence constraints. Hard-CAD's optional action module compares declared sequences and qualified precedence edges; it does not plan motions or inherit that completeness guarantee.

These studies evaluate features, edits and engineering requirements. Generation conformance, judge–human agreement and statement-to-entity reference accuracy are distinct evaluation targets. Hard-CAD jointly analyzes global material changes and identified local conditions across the complete recorded cohort, including failed or unavailable measurements. Its revision records support inspection of what a proposed repair changes and preserves; optional component and action modules are specified separately from this geometry evidence.

3. Task and Design Goals

The task maps an explicit geometric brief to executable CAD source and a solid model. Each case supplies a reference construction and supported requirements. Execution must yield an unambiguous artifact before checking. Both geometries use the declared frame and millimetres, without score-fitted registration.

Table 1. Documented scope and inputs. Section 6.5 compares the fixed-pose CADGenBench component; other rows summarize scope, not matched performance.

System	Reference or checking inputs	Relation to this study
AssemCAD	Typed parts, geometry-backed ports, mates and axioms	Grounded assembly realization and requirement traceability.
CADGenBench	Declared keep-in/keep-out volumes, surrounding-material shell and bounded pose search	Same-region fixed-pose component comparison; full benchmark score not run.
CADTests	Executable prompt-specific geometry/topology tests	Native requirement tests and feedback used in our transfer study.
CIT-CAD	Inferred entities, hierarchy, operations and relations	Construction-aware checks; repairs preserve satisfied constraints.
TraceCAD	Recovery state, localized search, repair memory and paired geometry	Fixed-frame volume analogue of preservation metrics; full system not reproduced.
CADEngBench	Parametric parts, body pairs, joint/analysis references	Parametric editing, engineering behavior and kinematic evaluation.
BenchCAD	Executable programs and geometric/feature references	Feature checks, edit preservation and high-IoU local defects.
Pointer-CAD v2	Dimensioned plans and reference B-Rep elements	Vertex, edge and face fidelity at the original scale and location.
Conventional CAD	Assembly components, clearance bounds and motion inputs	Established clearance, motion and interference operations.
Hard-CAD (evaluated)	Explicit brief/fixture requirements, bound operands and states	Paired global material changes and local requirement transitions, with failures retained.

Controlled constructions specify normal, violated and repaired states to isolate checking inputs. The agent cohort instead contains generated programs for 30 authored briefs specifying primitive dimensions, Boolean operations and, where specified, separate parts. It supports descriptive analysis of generated outputs, not an industrial-population estimate.

Implemented extensions. Optional component, synthesis and action adapters (historically T_1–T_6) are specified in Supplementary Appendix A. Drawing and process-layout adapters contribute no results. Component matching and action alignment need semantic references absent from this cohort; these modules do not establish a validated three-axis compliance scale.

4. Contract and Diagnostic Procedure

4.1. The contract

A contract records a requirement identifier (ID), source, entities, frame, units, state or path, comparison operator and bound. The ID anchors repeated measurements, the entities and frame locate the check, and the state and bound define its evaluation. Values and measurement status remain separate. A minimum gap needs a part pair and lower bound; path obstruction additionally needs the moving body, obstacles and trajectory.

Controlled contracts are author-declared. For the generated cohort, each brief specifies geometric primitives, their dimensions and placements, and the Boolean operations that combine or subtract them. We call these construction instructions the brief's recipe. Case, part and cut indices identify requirements across revisions. The released region interface rejects duplicate IDs and changed contract hashes before pairing reports (Supplementary Appendix B). Natural-document extraction is evaluated separately.

For example, the shaft–guide contract binds the shaft and guide in their installed state and requires a minimum gap of 0.05 mm. At a bore-axis offset of 0.12 mm, the nominal gap is 0.03 mm and the signed margin is −0.02 mm: the bound is violated despite positive clearance. The separate worst-size calculation gives a 0.01 mm gap and a −0.04 mm margin. Retaining the operand names, state and type of calculation makes these two results distinguishable (Section 6.1).

4.2. Input to measurement

The procedure qualifies inputs, obtains the candidate, executes applicable checks and reports requirements, operands, values and status. No declared cut means no applicable cut-region check; missing geometry, unresolved bindings or a missing path make a declared check unavailable. Neither is a measured zero. Revisions use the same reference and checking conditions.

The document adapter separates raw dimensions from millimetre values. Conversion requires explicit length units and a phrase identifying the mechanism's overall extent, rather than workpiece dimensions. This textual anchor identifies scope, not a CAD frame or entity. Force and mass remain separate; ambiguous cut/separation terms remain source mentions without inferred operations or geometric checks. Section 6.7 tests these corrections on real files. Component instances and action precedence still need supported references; the geometric studies use explicit briefs.

Table 2 states the input and interpretation of each evaluated check. Optional semantic modules are defined separately in Supplementary Appendix A.

5. Checks and Measurement Records

5.1. Notation and inputs

Let X be candidate occupied material and Y the reference, represented as unions of B-Rep solids in the declared common frame. $|\cdot|$ denotes volume in mm^3. Let b_i and $r_i > 0$ be candidate and required envelope dimensions in mm. A contract q identifies the quantity to measure, its operands and any required bound. Each output records $(q, \text{value}, \text{unit}, \text{status})$, retaining whether that check was applicable and successfully measured.

Table 2. Inputs and outputs of the evaluated checks. Every geometric operand must be valid and bound to the declared reference. Missing operands produce an unavailable result; zero denotes a measured value. Execution and artifact acquisition are prerequisites, not engineering acceptance.

Check	Contract/artifact inputs	Reported quantity and interpretation
Execution	Source, runtime and expected result object	Execution status and reason; D_{comp} where the adapter implements it.
Global overlap	X, Y, frame, unit, Boolean tolerance	Precision, recall, IoU; extra and missing material.
Envelope	Candidate bounds and required positive dimensions r_i	Returns 1 if $\max_i(\lvert b_i - r_i\rvert/r_i) \le 0.05$, otherwise 0; checks outer dimensions only.
Solid witnesses	Declared occurrences and a map to imported solid indices	Fraction w with valid injective witnesses; not semantic recognition.
Clearance	Named part pair, state and lower/upper gap bound	Distance and signed requirement margin in mm.
Prescribed motion	Moving solid, obstacle union, fixed path and state	Swept intersection volume in mm^3 for that path.
Declared cut region	Brief's cut primitive, additive body and final reference	Candidate occupancy of an explicitly required void region.

5.2. Occupied material and local conditions

Global precision, recall and intersection-over-union are

$$p_V = \frac{|X \cap Y|}{|X|}, \qquad r_V = \frac{|X \cap Y|}{|Y|}, \tag{1}$$

$$I_V = \frac{|X \cap Y|}{|X \cup Y|}. \tag{2}$$

The denominators must be positive. We also report $E_V = |X \setminus Y|$ and $M_V = |Y \setminus X|$. Recall alone cannot reject extra material: an enclosing solid has recall one. None of these quantities establishes that a particular hole, interface or component is correct. There is no global IoU acceptance threshold.

For a fixed reference volume $V_Y = |Y|$, the same quantities obey

$$I_V = \frac{V_Y - M_V}{V_Y + E_V}. \tag{3}$$

Writing Δ for feedback minus initial and using subscripts 0 and 1 for those endpoints gives

$$I_{V,1} - I_{V,0} = \frac{-\Delta M_V - I_{V,0}\Delta E_V}{V_Y + E_{V,1}}. \tag{4}$$

Thus an increase in extra material can accompany an increase in IoU when $-\Delta M_V > I_{V,0}\Delta E_V$.

The aggregate quantities retain volumes but not their locations. The diagnostic record therefore also retains the identified regions to which a change belongs. Supplementary Appendix B gives an elementary equal-volume redistribution example and its analytic regression fixtures.

For a lower-bound requirement $d \ge d_{\min}$, the signed margin is $d - d_{\min}$; for an upper bound it is $d_{\max} - d$. A negative margin denotes violation of that particular bound. Nominal geometry, worst-size bounds and signed margins are different quantities. Size intervals are designed bounds, not statistical confidence intervals.

A prescribed-motion check forms the swept occupied solid along the declared rigid path and intersects it with the specified installed obstacles. A zero result indicates that the declared rigid sweep is unobstructed in the specified state. Alternative-path search, deformation and contact mechanics are outside this check.

A witness binds one of $N > 0$ declared occurrences to a STEP solid index. Validity requires matching declared and actual artifact hashes, an existing index, a valid closed solid of positive volume, and no index reuse. Reused bindings all fail; no maximal injective subset is selected. With $z_i = 1$ for valid bindings and zero otherwise, $w = N^{-1}\sum_{i=1}^{N} z_i$: two reused bindings and one unique binding give $w = 1/3$. This tests declared correspondence, not semantic recognition or the absence of unreferenced bodies.

5.3. Numerical and execution conditions

Registration is declared, never fitted. The controlled fixtures use millimetres in a right-handed XYZ frame, with the base bottom at $z = 0$ and nominal shaft axis at $x = y = 0$. Solid Booleans use FreeCAD 1.1.1 and Open CASCADE 7.8.1. Each measurement binds the source, configuration, kernel and input hashes. The controlled geometry is checked at Boolean tolerances 10^{-6}, 10^{-5} and 10^{-4} mm; this measures numerical sensitivity, not manufacturing uncertainty. Figure tessellation does not enter these volumes.

Within-part material is unioned before comparison, avoiding double-counting overlapping solids. Ordinary assembly interference instead sums intersections over unordered actual-part pairs. A single enclosing solid has no such pair, so a zero pair sum is not evidence of a valid assembly.

Where defined, $D_{\mathrm{comp}} \in \{0, 1\}$ records successful source execution with the expected runtime result. Failure reasons retain syntax, import, timeout and output-selection distinctions. Missing execution prerequisites are recorded explicitly. Geometry is not assigned a score when no valid artifact is obtained, and the local agent study reports acquisition status rather than inferring D_{comp} from geometric similarity.

The generated-output analysis additionally checks the imported STEP artifacts before using their Boolean results. The replay and consistency criteria in Supplementary Appendix B determine which artifacts enter the Boolean-volume analysis. Functional checks use their own declared operands, states and bounds.

6. Evidence and Results

The studies test contract checks, recorded revision changes, feedback utility and document extraction. Table 3 summarizes their evaluation units and recorded results; the following subsections give the protocols and denominators.

6.1. Controlled geometry and sensitivity

The shaft–guide family comprises 21 bore offsets and four controls, including the original 11 constructions. Repeated measurements at three Boolean tolerances add no independent cases. Figure 1 consolidates the requirement, violation and repair examples; Table 4 gives representative distinctions. Detailed sweep values remain in the source-bound supporting results.

Analysis before enumeration. Let e be bore-axis offset, $R_b = 5.15$ mm the bore radius and $R_s = 5.00$ mm the shaft radius. Before contact, nominal clearance is

$$c(e) = R_b - R_s - e, \qquad \frac{dc}{de} = -1. \tag{5}$$

For $0 \le e < R_b - R_s$, reference and candidate occupied volumes remain equal to V. If $\Delta(e)$ is their symmetric-difference volume, then

$$I_V(e) = \frac{2V - \Delta(e)}{2V + \Delta(e)}, \qquad \frac{dI_V}{de} = \frac{-4V\Delta'(e)}{(2V + \Delta(e))^2}. \tag{6}$$

For the translated circular bore, $h = 20$ mm and the nominal occupied volume is $V \approx 91975.16$ mm^3. The symmetric difference is $\Delta(e) = 2h[\pi R_b^2 - A(R_b, e)]$, with circular-lens area

$$A(R, e) = 2R^2 \arccos\left(\frac{e}{2R}\right) - \frac{e}{2}\sqrt{4R^2 - e^2}. \tag{7}$$

The right derivative gives $I_V'(0) = -4R_b h/V \approx -0.00448$ mm^{-1}. An offset of 0.12 mm consequently leaves IoU about 0.99946 while reducing clearance to 0.03 mm, below the required 0.05 mm, with zero static interference. The unchanged-volume derivation is restricted to the noncontact interval; it is not applied to intersecting or missing-part controls.

Agreement between the kernel and analytical formula checks implementation consistency. Independent radius bounds of ±0.01 mm give a worst-size gap of 0.01 mm and a requirement margin of −0.04 mm; these are not the nominal gap.

An additional 27-pair redistribution fixture checks known volume identities and local-state reporting across three part families. Its outcomes are fixed by construction; Supplementary Appendix B retains the formulas and measured agreement as implementation tests.

Prescribed order. Cover-first insertion intersects the installed cover by approximately 201 mm^3 despite zero final-pose interference. Shaft-first repair removes that obstruction under the checked path. Identical final solids cannot encode installed-body state. This does not assert that every alternative assembly path is impossible, and it is not an action-sequence score.

6.2. Component-witness mutations

Three layouts (shaft–bushing, dowel–plate and slider–channel) each undergo seven declared mutations. All component labels are complete by construction. Table 5 therefore omits a constant label column and tests the bindings and geometry behind those declarations.

Fusion and duplicate binding both leave occupied-material IoU maximal while only one of three occurrences retains a valid injective witness. Duplicate binding corrupts correspondence without changing geometry; it is not evidence that an unreferenced body is absent. The offset mutation instead preserves bindings while causing interference. Repairs return the respective measures to their nominal states. These tests exercise deliberately injected defects, not observed model error frequencies.

6.3. Bounded carriage travel

A separate six-role carriage layout requires 18 mm of axial travel. Its normal, restricted and repaired constructions allow 20, 16 and 20 mm respectively. The restricted construction is initially collision-free over all 15 part pairs, but its requested endpoint and exact translation sweep both intersect the stop by 117.6 mm^3. Restoring the stop removes both intersections. The value is predicted by $(18 - 16) \times 9.8 \times 6$; kernel agreement tests the adapter, not an unexpected effect.

6.4. Complete local agent cohort

A headless adaptation of FreeCAD AI code mode [40] generated source for 30 original geometric briefs under Qwen2.5-Coder 7B and 14B conditions. The generator receives the modeling brief; the evaluator constructs the reference B-Rep from its recipe and computes the geometric feedback. The protocol fixes requested prompts, tools, temperature zero, seed and a 4096-token response budget. Each trajectory permits up to three responses to acquire a constructed artifact, then one geometric-feedback response. The 60 trajectories used 132 responses, yielding 50 initial and 49 feedback B-Reps. The 33 unmeasured attempts include execution or acquisition failures; they are not measured mechanical defects. Served context and actual token use were not captured per response, so the cohort does not support ranking the models.

Four archived cad11 endpoints have a volume-identity discrepancy above the existing 10^{-7} IoU audit limit. The revision also checks exported-artifact suitability, as detailed below. Every attempt and exclusion remains in the exported cohort table, and archive-only sensitivity analyses retain the original measurements and flags. Initial means the first constructed output, which may follow failed execution attempts; feedback comparisons use only matched eligible endpoints with the same reference.

Artifact replay and eligibility. The archived measurements use in-memory B-Reps rather than reimported STEP geometry. We verified those files against their recorded hashes and replayed their geometry without executing model code. Additional Boolean checks identify unsuitable geometry at 16 endpoints (eight reference and eight candidate imports), including all four cad11 numerical flags. These endpoints are unavailable for the primary analysis. Eligibility requires passing

Table 3. Completed studies and recorded results. Counts refer to constructions, documents or recorded trajectories, not independent industrial mechanisms. Geometric and extraction results are reported here; the optional semantic U/action modules are described separately in Supplementary Appendix A and are not evaluated by these studies.

Study	Evaluation units	Recorded result
Shaft–guide	25 constructions	At 0.12 mm offset: IoU 0.99946, gap 0.03 mm; shortfall 0.02 mm against the 0.05 mm minimum, zero interference.
Void redistribution	27 constructed pairs	Checks known volume identities and local-state reporting across three part families.
Witness mutations	3 layouts × 7 mutations	Injective solid binding, overlap and applicable interference under fixed complete labels.
Carriage	Normal/limited/repaired	Travel 20/16/20 mm against an 18 mm requirement; the limited case has 117.6 mm^3 endpoint/sweep intersection.
Recorded agent revisions	60 trajectories; 41 eligible pairs	84 paired requirements: one new and one cleared violation, localized alongside global material changes.
Real-document audit	22 documents; 13 explicit-unit triples	Literal precision 12/12, recall 12/13; metadata links 17 products in nine cases; feature semantics remain unqualified.
Proposal-to-checker replay	30 template briefs; 61 cut operations	Post-hoc decoding recovers 61/61 operations from unchanged responses; instantiated regions reproduce 169 local verdicts.
External interface component	77 endpoints; 169 regions	80/84 paired transitions agree with occupancy; interface preservation and empty-region checks differ on four.
Complete-recipe feedback	42 starts; 126 calls	All three arms repair the same 14 initially violated starts.
Bounded patch studies	42 starts; 84 calls each	Global/region joint repairs: 6/14 versus 7/14, then 2/14 versus 1/14.
Tool-using repairs	27 initial + 18 follow-up; 18 GLM extension	MiniMax-M3 and GLM: measured clearance, installation-path and travel repairs; final success ties across feedback conditions.
External transfer	50 released starts; 58 repair calls	Native CADTests: 1/29 aggregate versus 4/29 detailed-feedback joint successes; one observed new requirement failure.
Feedback pilot and repeats	164 pilot + 78 follow-up calls	Pilot A/B/C/D: 9/13, 9/13, 11/13, 8/13; repeated B/C: 30/39, 35/39 on 13 starts.

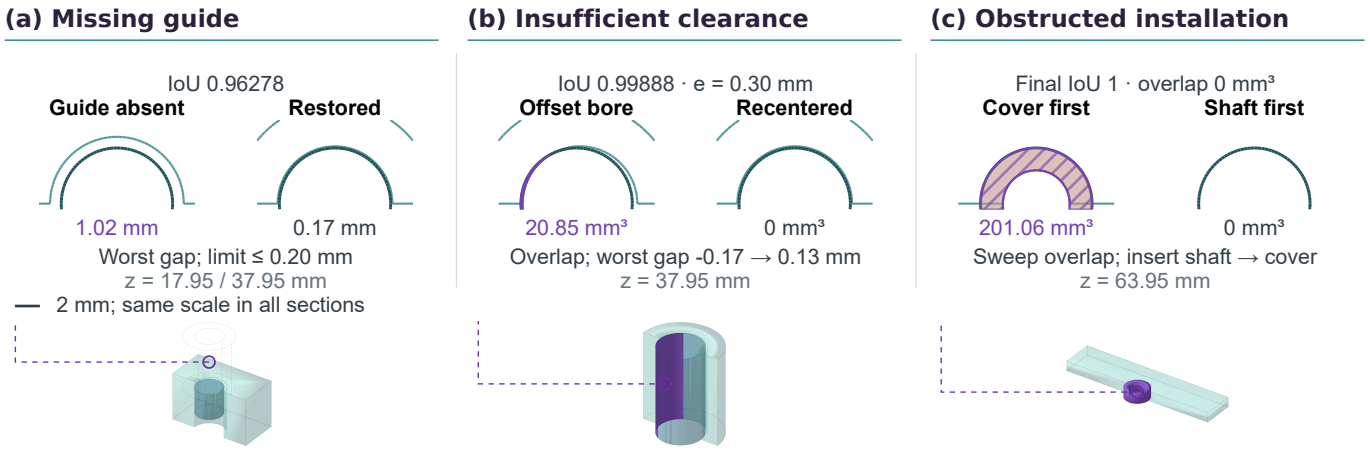


Fig. 1. Controlled requirement-to-repair examples: (a) a missing guide, (b) an offset bore and (c) an obstructed installation sweep. Small semi-transparent CAD views locate the checked region; enlarged views above show the upper halves of horizontal sections at the stated heights, before and after correction. Dark teal lines denote shaft or sweep boundaries, light teal lines denote supports, and violet hatching denotes intersection; the missing guide is dashed. Gap bounds use independent radius limits of ±0.01 mm, not confidence intervals; e is the bore-axis offset.

the geometry, reference-control, reference-reconstruction and replay-agreement checks detailed in Supplementary Appendix B. Excluded endpoints retain their archived measurements in sensitivity results, not primary scores.

Distributions and missingness. The primary analysis contains 83 eligible endpoints: 42 initial and 41 feedback. Ten trajectories never produce an initial B-Rep; one further trajectory has an initial B-Rep but an unmeasured feedback attempt. Table 6 reports each condition separately. Figure 2 shows all eligible pairs and two requirement-level case studies. The endpoint summaries describe this finite cohort conditional on eligibility; the conditions are not a model ranking.

The source tables additionally report E_V/V_Y, M_V/V_Y, candidate-minus-reference solid count, and the maximum absolute difference between corresponding fixed-frame bounding-box coordinates in mm. Across the 83 eligible endpoints, the median normalized extra volume is zero and its upper quartile

Table 4. Representative controlled conditions. The gap uses nominal geometry. "Final" and "sweep" refer to different states; zero means a performed intersection check. All volumes are mm^3.

Condition	Requirement-specific observation
Offset 0.12 mm	Gap 0.03 mm < 0.05 mm; static intersection 0; IoU 0.99946.
Cover installed first	Final intersection 0; prescribed insertion sweep about 201.
Shaft installed first	Same final solid; insertion and later-cover sweeps 0.
Single enclosing solid	Recall 1; precision/IoU 0.153; no part pairs.

Table 5. Witness mutations across three layouts. w is the fraction of declared occurrences with a valid injective solid witness; IoU compares occupied material. Ranges span layouts. Local interference is not measured when corrupted bindings leave operands undefined.

Mutation	w	IoU	Interference (mm^3)
Nominal	1.00	1	0
Missing body	0.67	0.843–0.956	not measured
Fused bodies	.33	1	not measured
Duplicate binding	.33	1	not measured
Interface offset	1.00	.99595–.99946	3.79–21.60
Enclosing solid	0	0.371–0.468	not measured
Repaired offset	1.00	1	0

is 0.0644; the median bounding-coordinate difference is zero and its upper quartile is 1 mm. These quantities preserve size and representation differences without assigning a general defect label. The paired table includes the change in normalized extra volume.

The 14B median IoU is 1.0 on 17 eligible endpoints per stage. At the 10^{-6} mm^3 threshold defined below, 15 have no extra or missing material; cad09 has 3500 mm^3 extra and cad18 has about 1206.37 mm^3 missing, with IoU 0.875887 and 0.887324 respectively. The 14B condition excludes five measured endpoints per stage; 7B excludes three. Eight 14B and two 7B trajectories lack an initial B-Rep, accounting for all ten missing initial outputs.

Global overlap and material discrepancy. Define a material discrepancy as extra or missing volume above 10^{-6} mm^3, a numerical screening threshold rather than a manufacturing tolerance. Among eligible endpoints with IoU at least 0.95, 0.98 and 0.99, respectively, 6/52, 4/50 and 2/48 retain such a discrepancy. At 0.95 each stage contributes 3/26; pooling stages does not create independent cases. These are conditional frequencies after artifact checks. The archive-only sensitivities in Supplementary Appendix B show how the denominator and frequency change when weaker eligibility rules are used.

Declared cut-region checks. The original briefs contain 61 declared cut primitives across 27 cases; three cases contain none. For cut primitive C_{ij} of part i, let A_i be the union of that part's additive primitives and Y the full reference-material union. The required void region and occupied portion are

$$Q_{ij} = (C_{ij} \cap A_i) \setminus Y, \qquad v_{ij} = |X \cap Q_{ij}|. \tag{8}$$

Table 6. Acquired geometry and IoU after eligibility checks. Each condition starts with 30 briefs. The 99 measured endpoints include 16 subsequently excluded by numerical-consistency and replay checks. IoU summaries use only the 83 eligible endpoints. All cohort quartiles use exclusive linear interpolation at sorted position $(n + 1)p$. Initial is the first constructed response; feedback follows initial construction.

Condition	Stage	Acquired	Eligible	IoU median [Q1,Q3]
7B	Initial	28	25	.862 [.630,1]
7B	Feedback	27	24	.862 [.695,1]
14B	Initial	22	17	1 [1,1]
14B	Feedback	22	17	1 [1,1]

Subtracting the full reference union prevents intended material of another part from being called a void violation. A positive-volume region is violated when $v_{ij} > 10^{-6}$ mm^3; absent or empty regions are not counted as passed checks. Since Q_{ij} lies outside Y, $0 \le v_{ij} \le E_V$: this check localizes extra material to a declared requirement, rather than detecting an error that an exact extra-volume check could not detect. Overlapping regions are reported separately and their occupancies are not summed into a global defect volume. Nominal-reference and deliberately filled-cut controls check zero and positive occupancy; reconstruction and occupancy-consistency tolerances are specified in Supplementary Appendix B.

Among the 83 eligible endpoints, 77 have applicable cut regions: 27 occupy at least one declared void and 50 do not; six have no applicable cut requirement. At IoU thresholds 0.95, 0.98 and 0.99, the counts are 6/48, 4/46 and 2/44 among eligible endpoints with applicable regions. Four high-IoU endpoints have no cut requirement: hence 6/48 here versus 6/52 for material discrepancy at 0.95. The six violations at 0.95 are the initial and feedback outputs of three 7B trajectories (cad02, cad03 and cad25); each pair retains the same program. They are three repeated-case examples, not six independent failures. Thus high overlap can coexist with violation of an explicit void requirement on generated outputs.

Paired feedback. Of 60 trajectories, 41 have two eligible endpoints, eight are excluded by artifact/numerical checks and eleven lack a measured pair. An increase requires dimensionless $\Delta I_V > 10^{-6}$ for IoU or $\Delta E_V > 10^{-6}$ mm^3 for extra volume. IoU increases in 3/41 pairs; extra material also increases in two of those three, or 2/41 of all eligible pairs. Both events occur in the 7B condition (cad06 and cad30). Thirty-four pairs retain identical program hashes and zero material changes; four further changed programs leave these volumes unchanged. The near-origin mass in Figure 2 therefore reflects repeated or volumetrically unchanged outputs, not 38 independent demonstrations of successful repair.

Requirement-wise changes. Of the 41 eligible pairs, 38 have applicable cut regions and three have none. Matching the same case, part and cut across endpoints gives 84 paired requirement instances across 25 briefs: 63 remain unoccupied, 19 remain violated, one becomes violated and one is cleared. Regions and model conditions are repeated measurements, not independent cases. In cad15/7B, IoU increases from 0.229885 to 0.230011 and extra material falls by 28.27 mm^3, with no missing material

at either endpoint, yet a previously clear side-cut region acquires 28.27 mm^3 of material. The program shifts the cut's starting coordinate from $x = -26$ to -24 mm. Conversely, cad30/7B clears 37.70 mm^3 from its declared void while total extra material increases by 1987.30 mm^3. These are the sole new and resolved cut violations in the complete applicable paired cohort. The paired records locate the new obstruction in cad15 and the cleared region in cad30, while the global quantities describe each edit's net material change. Section 6.6 separately tests how diagnostic feedback affects repair outcomes.

Edit decomposition and coordinate/scale checks. We further replay all 41 pairs to locate the material additions and removals behind these changes. For initial and revised solids X_0, X_1, let $A = X_1 \setminus X_0$ and $D = X_0 \setminus X_1$. Established Boolean differences, also used for CAD debugging [22], give $\Delta E_V = |A \setminus Y| - |D \setminus Y|$ and $\Delta M_V = |D \cap Y| - |A \cap Y|$. In cad15, 56.55 mm^3 is removed from one declared void while 28.27 mm^3 is added to the other. Both volumes lie outside the reference, giving a net extra-volume reduction of 28.27 mm^3 when computed before rounding. Even occupancy of the union of all declared voids decreases while that individual requirement regresses. In cad30, the revision adds 675 mm^3 of required material and 2025 mm^3 outside the reference, while clearing 37.70 mm^3 from the required void. The cad30 source identifies a cover raised from $z = 0$ to 30 mm while retaining the incorrect (x, y) origin $(0, 0)$ instead of $(-15, -15)$ mm. Only its $15 \times 15 \times 3$ mm overlap fills the intended cover; the remaining three quarters lie outside the reference. Removing its old position clears a quarter-cylinder of radius 4 mm and height 3 mm from the base void. The decomposition attributes the material changes to the misplaced cover and the base void; assembly function would require an additional check of their intended interaction.

Coordinate/scale checks and a second OCCT version reproduce these measurements on the same archived solids. They test numerical consistency; the observational sample remains 41 pairs. Supplementary Appendix B gives the transformations, tolerances and execution record.

Interpreting the two paired events. In cad06/7B, feedback restores a mistakenly subtracted ring through a Boolean union. IoU rises from 0.167 to 0.686, extra material from about 3848 to 4445 mm^3, and missing material falls by about 7461 mm^3. In cad30/7B, missing material falls by 675 mm^3 while extra material rises by about 1987 mm^3 and IoU rises from 0.163 to 0.194. Equation 4 explains both: cad30 needs a missing-material reduction above only $0.163 \times 1987 \approx 324$ mm^3 for IoU to increase. Global improvement need not reduce every discrepancy.

6.5. Comparisons with existing measurements

We ran CADGenBench's fixed-pose keep-out interface component [34] on the same archived candidates, references and declared forbidden regions. All 77 applicable endpoints contribute their 169 region instances; six no-cut endpoints remain not applicable. This uses the upstream `iou_at_pose` function, its 0.95 pass threshold and 0.99 saturation threshold. It omits the full benchmark's alignment, pose search and group scoring. The external criterion also checks surrounding reference material, whereas declared void occupancy only asks whether the specified region is empty.

Both criteria pass 42 endpoints and fail 27; eight pass the occupancy check but fail the external interface criterion. No endpoint shows the reverse disagreement. All 169 regions are measured: 121 pass both, 40 fail both and eight disagree. Uniformly halving the mesh deflection changes no region or endpoint verdict; the largest continuous-score change is about 0.0021. This is agreement between two defined criteria, not accuracy against independent functional truth.

All eight disagreeing endpoints, spanning five briefs, have missing intended material around the declared void in a separate B-Rep check. A simple enlarged-hole control illustrates the distinction: the required region is empty, but excess removal reduces external interface IoU to 0.25. Nominal and filled-hole controls produce interface IoU 1 and 0. The disagreement has a geometric explanation: occupancy checks whether the required void is empty, whereas the interface criterion also penalizes removal of surrounding reference material. The paired record retains the chosen criterion and requirement identity, making clear which condition changed.

Simple difference baseline. To test whether transition tracking requires a distinct checking method, we joined the external region verdicts by case, model, attempt and requirement identity, verified candidate and reference hashes, and differenced the two stages. All 84 paired requirements were retained. The external criterion gives 60 clear-to-clear, 23 violated-to-violated and one new violation; occupancy gives 63, 19 and one, plus one cleared violation. The transition labels agree in 80/84 instances. The four differences are three regions that remain empty while failing interface preservation at both stages, and one occupancy repair that still fails interface preservation. Existing checker outputs therefore integrate directly into requirement-wise change records. The four disagreements arise from the acceptance criteria, rather than from the pairing operation.

Comparison with preservation metrics. TraceCAD already measures retained correct geometry and newly introduced false geometry across a repair [24]. Applying a fixed-frame continuous-volume analogue of those definitions to the archived Boolean partitions, all 41 pairs retain their initial reference-overlapping material. Three pairs introduce new material outside the reference: cad06, cad15 and cad30, all in the 7B condition, with normalized new-false-geometry values approximately 0.0614, 0.00239 and 0.298. Thus the continuous-volume analogue of that preservation criterion also identifies additions in both highlighted local cases, including cad15 despite its falling net extra volume. The requirement records identify which declared region changes state. Supplementary Appendix B gives the analogue and its differences from TraceCAD's aligned voxel implementation; this analysis adds no generated samples.

6.6. Fixed-budget feedback comparison

The complete-recipe study used 42 initial endpoints; all arms cleared the same 14 violated starts (126 calls). Two bounded-patch studies (84 calls each) compared global feedback with

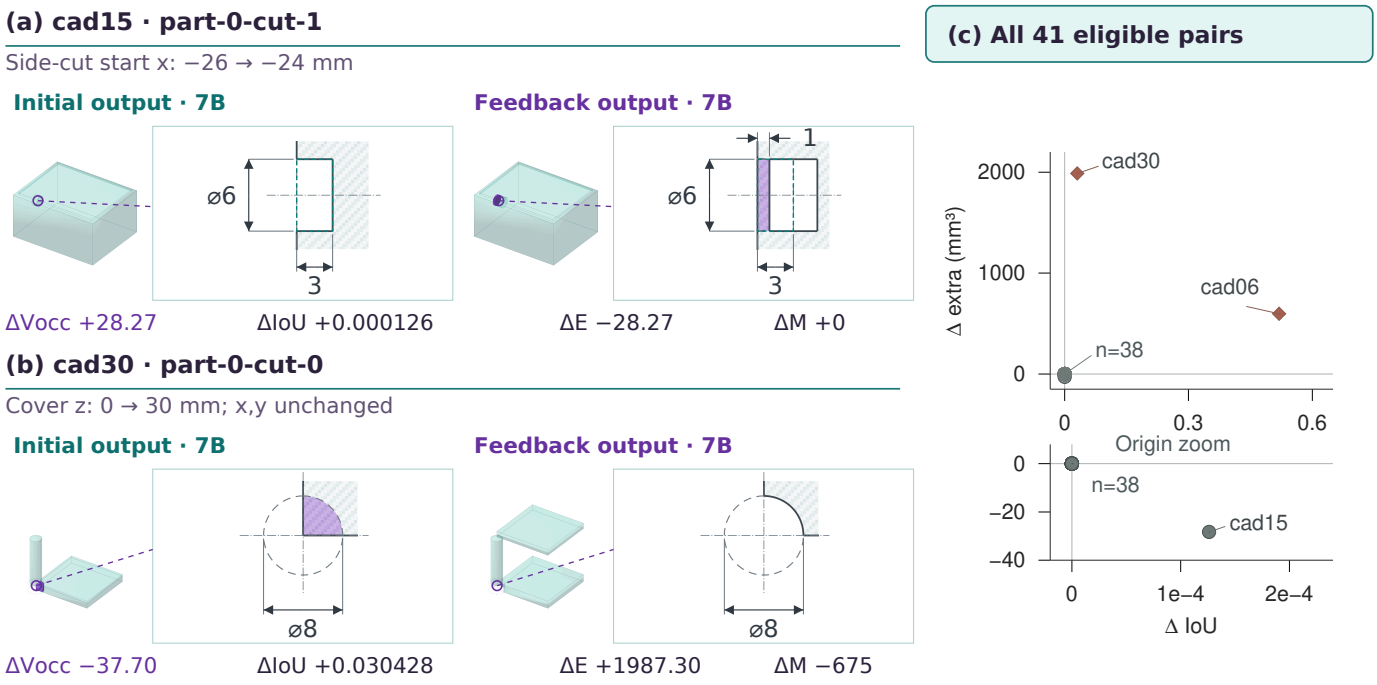


Fig. 2. Material changes and matched local requirements. (a) cad15 gains a 1 mm web inside the declared 6 mm diameter side cut, although IoU improves and extra material falls. (b) cad30 clears the quarter-circle obstruction in its base cut while extra material grows. Within each case, translucent CAD views share a camera and scale; adjacent dimensioned sections show A–A ($y = 0$) and B–B ($z = 1.5$). Hatching denotes sectioned material; violet marks material inside the required void. The dashed teal contour fixes the cad15 requirement region. Labels identify initial and feedback-generated 7B outputs and their parameter edit. The change records report occupancy (ΔV_{occ}), IoU, extra material (ΔE) and missing material (ΔM); all volumes are in mm^3. (c) All 41 eligible pairs on linear axes, with an origin enlargement showing the 38 coincident pairs and cad15. Diamonds mark simultaneous IoU and extra-material increases (cad06/cad30). The two local cases represent one new and one cleared violation among 84 paired requirements.

added region evidence. Joint-repair counts were 6/14 versus 7/14 with the full program visible, and 2/14 versus 1/14 with a five-line fragment. Supplementary Appendices B and C distinguish starting cohorts, success criteria, side-effect denominators and actual token costs.

Matched feedback pilot and repeats. A four-arm pilot supplied all 41 eligible feedback endpoints to the same locally served GLM model, with current code and declared cut task cards. Each arm allowed one full-program response and a 4096-token output cap. A received execution/preflight information; B added global geometry; C added current region states; D also received earlier region states and changes. The joint endpoint requires all targets to clear, all initially clear cuts to remain clear, and neither extra nor missing material to increase beyond 10^{-6} mm^3. Among 13 target-bearing starts, A/B/C/D achieved 9/13, 9/13, 11/13 and 8/13 joint successes; each produced 37/41 measurable candidates. No arm introduced an observed non-target cut violation. Unavailable outcomes remain unknown for side effects and failures for the joint endpoint.

Only two starts had changed regional history, one with a violated target. A frozen B/C follow-up on all 13 target-bearing starts used three responses per condition at temperature 0.3, with matched call/output caps and requested seeds. Joint successes were 30/39 and 35/39, with 38/39 measurable outputs per arm. C improved the per-start count on four cases, worsened it on one and tied on eight. The mean within-start difference was 12.8 percentage points; the predeclared two-sided sign-flip test on these magnitudes gave $p = 0.3125$. This descriptive advantage remains inconclusive at the case level, and neither arm introduced a measured non-target violation.

External transfer. We selected 50 of CADTestBench's 200 detailed-prompt cases by a frozen hash rule and executed native tests on the released GPT-5.2 programs [26]. All executed: 21 passed every test and 29 had both passing and failing requirements. Each of these 29 starts received one GLM repair per feedback condition, with the same starting code, task prompt, one-call budget and 4096-token cap. Aggregate feedback supplied passed/total counts; detailed feedback added per-test descriptions, statuses and messages. Joint successes were 1/29 and 4/29, with 22 and 20 fully measured revisions. Three paired successes favored detailed feedback, none favored aggregate feedback ($p = 0.25$, two-sided exact McNemar test). One detailed-feedback repair fixed a face-area check but raised the solid-to-box volume ratio from 0.118 to 0.955, violating the native L-profile threshold of 0.95. The solid retained a notch. This is a regression on disclosed native tests, not an IoU-improvement event or a GPT-5.2 repair.

Persistent-state development pilot. On eight previously inspected, hash-selected external starts, a shared accepted-state controller allowed three GLM patch rounds per arm. Current-check feedback achieved 3/8 joint successes and added history 5/8 (three paired wins, one loss, four ties). The same gate rejected all four proposed preservation regressions. These 16 tra-

jectories and 35 calls are development evidence, not a held-out estimate: first-round prompts differ and serving is not demonstrably deterministic. None of these studies reproduces TraceCAD's complete repair system [24]; a general repair advantage remains unestablished.

Tool-using agent repairs. MiniMax-M3 repairs three controlled assemblies through Hermes source-reading/writing tools and three history-retaining revisions. All feedback conditions reach full satisfaction on all tasks across 27 initial revisions. An information-matched follow-up compares complete path/value logs with the same measurements as JSON: both solve each task at the first revision and preserve success across 18 revisions. The first structured edits change clearance from 0.030 to 0.120 mm, swept intersection from 201.06 to 0 mm^3, and travel from 16 to 18 mm. All 45 grades reproduce in API-free CAD replay. A GLM-5.3-Flash-EXL3 service extension repeats the matched protocol: both arms solve each task at the first revision, and all 18 grades replay exactly. These three-task trials test edit/check/replay integration without separating the feedback formats' repair performance. Supplementary Appendix C retains controller details, incomplete turns and measured edits.

6.7. Real-document source audit

A retrospective audit verified 22 original DOCX files and their same-numbered STEP assemblies and design archives. A separate OOXML scan enumerated 21 labelled dimension triples: 13 with explicit units and eight without. The revised extractor recovered 12/13 explicit-unit triples with no unmatched output (precision 12/12, recall 12/13); the missed triple lacked the required envelope anchor. These targets measure literal recovery on the same cohort, not semantic binding or unseen-document performance. Rights-controlled sources remain private; Supplementary Appendices B and D retain the full protocols and historical corrections.

Matching the 13 explicit-unit triples to whole-assembly boxes yielded 0/13 matches under fixed XYZ at 1%, 5% and 10% error bands; six axis permutations yielded 2/13, 4/13 and 8/13. Individual-solid search produced only one numerical candidate at the widest band, without an authored entity link. Numerical compatibility therefore did not qualify the intended geometric references.

Engineering consultation informed the contract inputs: the audited joint response requested a required-component inventory and treated unspecified assembly dependencies as underdetermined. Explicit entities, frames and conditions make checks reproducible; their correspondence to the source still needs evidence.

Source-backed product links. Fifteen retrieved manufacturer case pages provide 33 explicit part-number rows; exact identifiers link 17 to unique STEP product definitions in nine of the 22 cases. All 17 pass the OCP/FreeCAD volume replay. Six definitions have multiple placed occurrences; a separate source-version screen finds one conflicting dimension signature. These executable links retain occurrence and version distinctions rather than infer a feature or missing condition (Supplementary Appendix B).

Candidate feasibility and binding. Across all 22 documents, a four-proposal cap with original STEP product names and no measured geometry yielded 20 literal/ID-valid proposals in 14 documents, five geometrically compatible in three documents. A separate source-indexed interface fixed values, units and text spans before product/feature selection: 37 proposals were admitted in 17 documents, eight compatible in four documents. These counts measure candidate availability under capped budgets, not semantic accuracy. Product identity, feature role and frame still require independent evidence.

For controlled integration, the 30 original recipes rendered as template prose supplied explicit bindings. Correcting only outer Markdown-fence parsing recovered 61/61 cuts from unchanged responses; instantiated regions reproduced 169 verdicts on 77 applicable endpoints. Supplementary Appendix B retains parser failures and excluded endpoints.

On 24 published BRepGround queries and STEP models [41], GLM selected the exact labelled face/edge set in 9/24 cases. Eight label-disagreeing selections still produced valid edits; all seven pairs with valid edits on both selections differed geometrically. This numeric-catalog pilot measures agreement with external labels, retaining view-dependent ambiguity. Successful execution alone leaves target identity unresolved.

Executable diagnostic report. Figure 3 traces the cad15 contract through FreeCAD measurements to the paired diagnostic record. Executable cad15 and cad30 examples produce JSON and HTML reports with the same requirement IDs, numerical changes and unavailable states.

7. Discussion and Limitations

7.1. Interpreting a revision

The controlled checks measure intersection, clearance and prescribed motion. Comparing each requirement across two versions distinguishes an existing defect from one introduced by the edit. The preservation-metric analogue also detects the added erroneous geometry in cad15/cad30. The requirement record identifies which specified region it affects.

Incorrect inputs can change more than the final verdict. All four wrong-entity controls change verdicts, whereas interpreting SR as spherical radius makes two radius fields checkable without resolving their semantic binding. Recording entities, frames, units and measurement status keeps these outcomes distinct. Signed margins also show changes that leave a verdict unchanged; we have not measured their sensitivity to binding errors in natural documents.

7.2. Evaluation scope

The 30-brief cohort contains cad15/cad30 transitions in the 7B condition. The 14B, GPT-5.2, GLM and MiniMax studies retain separate protocols and denominators. Prevalence needs representative sampling. Failures and exclusions remain explicit; six high-IoU void violations belong to three trajectories. Coordinate/kernel replays reuse shapes; repeated feedback varies responses within fixed budgets.

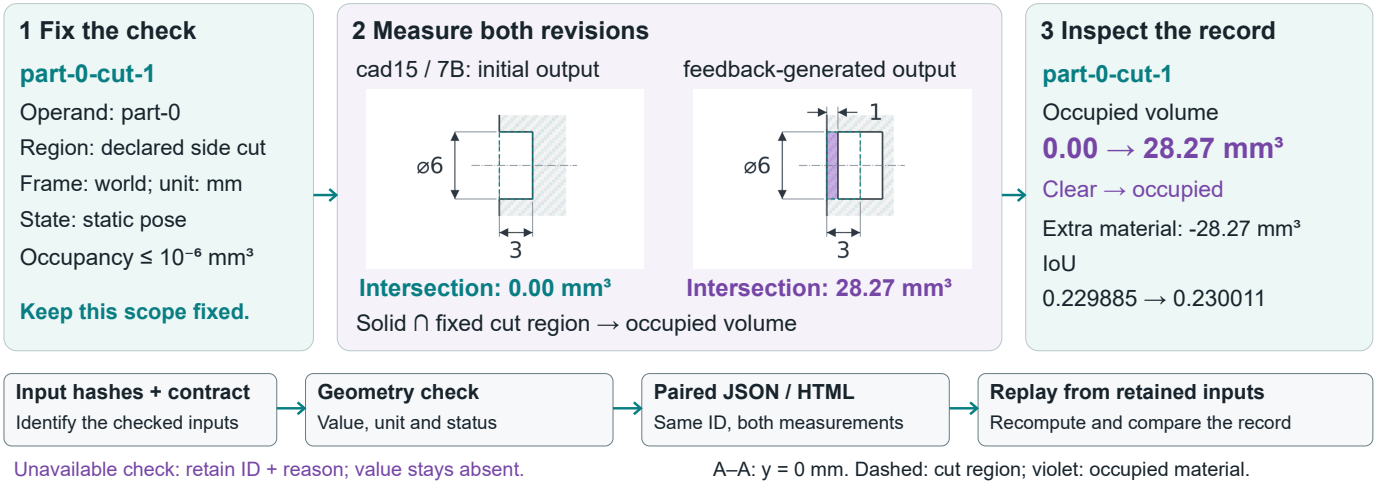


Fig. 3. Inspecting a revision with Hard-CAD. (1) Fix the operand, region, frame, unit, state and bound. (2) Intersect both candidates with that region; the cad15 sections locate the added material. (3) Pair measurements by requirement ID, retaining local occupancy and global changes. Archived CAD views and SDK reports supply the geometry and values. The lower path supports replay; its unavailable-check rule is not an additional cad15 outcome.

The document studies test extraction, geometric compatibility and exact product identifiers. Feature-level accuracy additionally needs the intended occurrence, feature and frame; the primitive pilot measures agreement with partly view-dependent labels. Current contracts cover explicit regions and bounded rigid translations. Rotation, deformation, loads and manufacturing need further state models and checks.

7.3. Practical use and next tests

Existing interface-checker outputs can be paired by requirement identity, but the report must distinguish surrounding-material preservation from empty-region occupancy. Added history yields fewer successes in the single-response pilot and more in the persistent-state pilot. Different cohorts, interfaces and budgets prevent attributing this contrast to the controller or claiming a general repair advantage.

Binding needs the placed occurrence, feature and state. Volume and diameter are rotation-invariant for an identified solid or circular feature; position and motion also need a frame or path. Held-out native constraints or semantic PMI could test binding errors. Repair comparisons should share the first proposal, branch once history exists, and fix the controller, budget and checks. Matched ordinary history isolates representation; withheld checks test preservation beyond disclosed requirements.

8. Conclusion

Hard-CAD pairs declared requirements across revisions alongside global material measures. In the eligible cohort, IoU improves despite a new local violation; another revision clears a void while extra material grows. Boolean differences locate the changes, and requirement identities distinguish repairs from regressions.

Diagnosis alone does not ensure better repair: interface-checker outputs support similar records, and added history has mixed results. Document studies distinguish dimension recovery from entity identification. Released protocols retain measured changes and unavailable results for further repair and binding tests.

Supplementary material and data availability

The independent supplement contains optional semantic definitions (Appendix A), reproduction protocols and additional results (Appendix B), and full feedback and document-study protocols (Appendices C and D). Executable checks, original fixtures and derived results form the reproducibility package; rights-controlled engineering inputs and raw provider captures are excluded.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

OpenAI Codex, Anthropic Claude and a locally served GLM coding assistant were used to assist literature searches, manuscript revision, code development and checks against submission requirements. CAD kernels compute the reported geometric quantities. AI tools were not used as independent experts or as ground truth for the reported engineering checks. The authors remain responsible for the scientific content and the decision to submit.

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